

Module 3 SQL Analysis Results

Questions 1–9 and the two original database questions.

Question 1: How many entries are from applicants who applied for Fall 2026?

Result

Fall 2026 applicant count: 32,600

SQL query

```
SELECT COUNT(*) FROM applicants WHERE term ILIKE '%Fall 2026%'
```

Explanation: Counts all rows whose term contains Fall 2026, regardless of other missing fields.

Question 2: Among classified entries, what percentage are international students?

Result

Percent international: 48.92%

SQL query

```
SELECT 100.0 * COUNT(*) FILTER ( WHERE LOWER(TRIM(us_or_international)) = 'international' ) / NULLIF(COUNT(*), 0)  
FROM applicants WHERE us_or_international IS NOT NULL AND TRIM(us_or_international) <> ''
```

Explanation: Uses classified rows as the denominator and counts only International rows in the numerator.

Question 3: What are the separate average GPA and GRE scores for applicants who provide each metric?

Result

Average GPA: 3.77
Average GRE Quantitative: 165.38
Average GRE Verbal: 160.18
Average GRE Analytical Writing: 4.35

SQL query

```
SELECT AVG(gpa), AVG(gre), AVG(gre_v), AVG(gre_aw) FROM applicants
```

Explanation: Calculates each metric independently, so missing GPA or GRE values do not exclude a row from another average.

Question 4: What is the average GPA of American applicants who applied for Fall 2026?

Result

Fall 2026 American average GPA: 3.79

SQL query

```
SELECT AVG(gpa) FROM applicants WHERE term ILIKE '%Fall 2026%' AND LOWER(TRIM(us_or_international)) = 'american' AND gpa IS NOT NULL
```

Explanation: Filters to Fall 2026 American applicants and averages only non-NULL GPA values.

Question 5: What percentage of Fall 2025 entries are acceptances?

Result

Fall 2025 acceptance percentage: 37.01%

SQL query

```
SELECT 100.0 * COUNT(*) FILTER ( WHERE LOWER(TRIM(status)) LIKE '%accept%' OR LOWER(TRIM(status)) = 'admitted' )  
/ NULLIF(COUNT(*), 0) FROM applicants WHERE term ILIKE '%Fall 2025%'
```

Explanation: Divides accepted Fall 2025 rows by all Fall 2025 rows, including rows without other optional fields.

Question 6: What is the average GPA of accepted applicants who applied for Fall 2026?

Result

Fall 2026 accepted average GPA: 3.78

SQL query

```
SELECT AVG(gpa) FROM applicants WHERE term ILIKE '%Fall 2026%' AND ( LOWER(TRIM(status)) LIKE '%accept%' OR LOWER(TRIM(status)) = 'admitted' ) AND gpa IS NOT NULL
```

Explanation: Filters to Fall 2026 accepted applicants and averages their available GPA values.

Question 7: How many entries are for a master's degree in Computer Science at Johns Hopkins University?

Result

Johns Hopkins master's Computer Science count: 31

SQL query

```
SELECT COUNT(*) FROM applicants WHERE LOWER(COALESCE(university, '')) LIKE ANY (ARRAY['%johns hopkins%', '%jhu%']) AND LOWER(COALESCE(program, '')) LIKE '%computer science%' AND LOWER(COALESCE(degree, '')) ~ '^(^[^a-z])(masters?|m\.?s\.?)(^[^a-z]|$)'
```

Explanation: Matches the original university, program, and degree fields using case-insensitive text patterns.

Question 8: How many qualifying Fall 2026 PhD Computer Science acceptances use the original fields?

Result

Question 8 count: 30

SQL query

```
SELECT COUNT(*) FROM applicants WHERE term ILIKE '%Fall 2026%' AND ( LOWER(TRIM(status)) LIKE '%accept%' OR LOWER(TRIM(status)) = 'admitted' ) AND LOWER(COALESCE(degree, '')) LIKE '%phd%' AND LOWER(COALESCE(program, '')) LIKE '%computer science%' AND ( LOWER(COALESCE(university, '')) LIKE '%georgetown%' OR LOWER(COALESCE(university, '')) LIKE '%massachusetts institute of technology%' OR LOWER(TRIM(COALESCE(university, ''))) = 'mit' OR LOWER(COALESCE(university, '')) LIKE '%stanford%' OR LOWER(COALESCE(university, '')) LIKE '%carnegie mellon%' )
```

Explanation: Uses the original university and program fields while also requiring Fall 2026, acceptance, PhD, and Computer Science.

Question 9: How many qualifying Fall 2026 PhD Computer Science acceptances use the LLM-generated fields?

Result

Question 9 count: 30

SQL query

```
SELECT COUNT(*) FROM applicants WHERE term ILIKE '%Fall 2026%' AND ( LOWER(TRIM(status)) LIKE '%accept%' OR LOWER(TRIM(status)) = 'admitted' ) AND LOWER(COALESCE(degree, '')) LIKE '%phd%' AND LOWER(COALESCE(llm_generated_program, '')) LIKE '%computer science%' AND ( LOWER(COALESCE(llm_generated_university, '')) LIKE '%georgetown%' OR LOWER(COALESCE(llm_generated_university, '')) LIKE '%massachusetts institute of technology%' OR LOWER(TRIM(COALESCE(llm_generated_university, ''))) = 'mit' OR LOWER(COALESCE(llm_generated_university, '')) LIKE '%stanford%' OR LOWER(COALESCE(llm_generated_university, '')) LIKE '%carnegie mellon%' )
```

Explanation: Uses the LLM-generated university and program fields while retaining the original term, status, and degree filters.

Question 10: Which five universities have the most Fall 2026 applicant entries?

Result

Stanford University: 735
University of California, Berkeley: 614
Yale University: 596
University of Washington: 590
University of Michigan - Ann Arbor: 566

SQL query

```
SELECT university, COUNT(*) AS applicant_count FROM applicants WHERE term ILIKE '%Fall 2026%' AND university IS NOT NULL AND TRIM(university) <> '' GROUP BY university ORDER BY applicant_count DESC, university ASC LIMIT 5
```

Explanation: Groups Fall 2026 rows by university, counts each group, sorts from largest to smallest, and returns five rows.

Question 11: What is the average GPA for American and international applicants who provide a GPA?

Result

American average GPA: 3.78
International average GPA: 3.77

SQL query

```
SELECT LOWER(TRIM(us_or_international)) AS classification, AVG(gpa) FROM applicants WHERE  
LOWER(TRIM(us_or_international)) IN ('american', 'international') AND gpa IS NOT NULL GROUP BY  
LOWER(TRIM(us_or_international)) ORDER BY classification
```

Explanation: Groups American and International rows separately and averages only rows that provide a GPA.